12.3 What Does "Random" Mean?

WORKSHEET

The question

(i) The point A lies on the circumference of a circle of radius a and centre O. The point B is chosen at random on the circumference, so that the angle AOB has a uniform distribution on $[0, 2\pi]$. Find the expected length of the chord AB.

(ii) The point C is chosen at random in the interior of a circle of radius a and centre O, so that the probability that it lies in any given region is proportional to the area of the region. The random variable R is defined as the distance between C and O. Find the probability density function of R.

Obtain a formula in terms of a, R and t for the length of a chord through C that makes an acute angle of t with OC.

Show that as C varies (with t fixed), the expected length $L(t)$ of such chords is given by

$$L(t) = \frac{4a(1-\cos^3 t)}{3\sin^2 t}.$$

Show further that

$$L(t) = \frac{4a}{3}\left(\cos t + \tfrac{1}{2}\sec^2\left(\tfrac{1}{2}t\right)\right).$$

(iii) The random variable T is uniformly distributed on $[0, \frac{1}{2}\pi]$. Find the expected value of $L(T)$.

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps.

Ask yourself If the angle AOB is θ, what is the length of the chord AB? (Drop a perpendicular from O.)

STAGE 2 A random point inside

Ask yourself What is $\mathrm{P}(R \le r)$ for $0 \le r \le a$? Differentiate to get the density.

Ask yourself A chord through C makes angle t with OC. How far is the chord from O?

STAGE 3 Averaging over C

Ask yourself Factorise $1 - \cos^3 t$ and $\sin^2 t = 1 - \cos^2 t$. What cancels?

Try this next

1. **Bertrand's probabilities.** Verify the three probabilities $\frac{1}{3}$, $\frac{1}{2}$, $\frac{1}{4}$ above.
2. **The other mean.** Show that if the chord's midpoint is chosen uniformly in the disc (the right-hand picture), the expected length is $\frac{4a}{3}$. (*Hint:* this is $L(\frac{\pi}{2})$. Why?)
3. **Random distance.** If the distance of the chord from O is uniform on $[0, a]$, show that the expected length is $\frac{\pi a}{2}$.
4. **Three methods, three distributions.** For each of the three methods, find the probability density of the chord's length ℓ on $[0, 2a]$.

Further reading

- **Edwin Jaynes, "The well-posed problem"**, *Foundations of Physics* 3 (1973). The invariance argument, readable and provocative.
- **Gábor Székely, *Paradoxes in Probability Theory and Mathematical Statistics*** (Reidel). Bertrand's paradox and many others.
- **Daniel Klain & Gian-Carlo Rota, *Introduction to Geometric Probability*** (Cambridge University Press). Integral geometry, for later.

The full guided solution, and what the question leads to at university, is in Section 12.3:
https://beyondstep.keithmcnulty.org/ch12/bertrand.html